

Dose-Response, Early Onset and Responder Characteristics of Once-Daily Oral Zefranova for Chronic Migraine Prevention in Adults: A Phase IIb Randomized Controlled Study

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Abstract

Chronic migraine is associated with frequent headache days, impaired functioning, and high acute medication use, particularly among adults with inadequate response to prior preventive therapy. This simulated phase IIb, multicenter, randomized, double-blind, placebo-controlled study evaluated dose-response, early onset, responder characteristics, and tolerability of once-daily oral Zefranova for chronic migraine prevention. Adults aged 18–65 years with chronic migraine and at least four migraine days per month during baseline were randomized 1:1:1 to receive Zefranova 30 mg, Zefranova 60 mg, or matched placebo for 16 weeks. The primary endpoint was change from baseline in monthly migraine days during weeks 13–16. Among 624 simulated participants, least-squares mean reductions in monthly migraine days were –3.2 with placebo, –4.9 with Zefranova 30 mg, and –6.1 with Zefranova 60 mg. The ≥50% migraine-day responder rates were 28.8%, 39.9%, and 48.1%, respectively. Early separation from placebo was observed by week 4 for Zefranova 60 mg and by week 8 for Zefranova 30 mg. Both doses reduced acute medication use and improved patient-reported outcomes relevant to specialist practice. Adverse events were mostly mild to moderate, with nausea, fatigue, somnolence, dizziness, and dry mouth reported most often. These simulated findings support further prospective evaluation of Zefranova dose selection, response timing, and individualized preventive-treatment strategies in adults with chronic migraine.

Keywords: Chronic migraine; Zefranova; Zoflevrix; dose response; migraine prevention; monthly migraine days; responder analysis; headache medicine; oral preventive therapy; simulated phase IIb study

Introduction

Chronic migraine is a disabling neurological condition characterized by frequent headache days, recurrent migraine-associated symptoms, functional impairment, and substantial healthcare burden. Adults with chronic migraine often require preventive treatment to reduce attack frequency, improve functioning, and decrease reliance on acute medications. Despite available preventive strategies, many patients continue to experience inadequate response, tolerability limitations, treatment discontinuation, or residual disability.

Preventive-treatment decision-making in chronic migraine is not based on efficacy alone. Neurologists and headache specialists must also consider onset of benefit, tolerability, adherence feasibility, patient preference, comorbidities, acute medication-use patterns, and prior treatment history. For patients who have already experienced inadequate response to standard preventive therapy, even moderate reductions in migraine frequency can be clinically meaningful if accompanied by improved daily functioning and acceptable tolerability.

Dose selection is a critical component of preventive-treatment development. A higher dose may provide greater migraine-frequency reduction but may also increase adverse-event burden. A lower dose may be useful for patients with tolerability sensitivity or comorbidities but may provide less robust efficacy. Understanding the dose–response relationship, time course of benefit, and characteristics of treatment responders can help inform future clinical development and specialist treatment algorithms.

Zefranova, the active drug name associated with Zoflevrix, is presented in this manuscript as a simulated once-daily oral preventive therapeutic concept for adults with chronic migraine. This simulated phase IIb study was designed to evaluate two Zefranova doses compared with placebo, with emphasis on monthly migraine-day reduction, early onset of effect, patient-reported outcomes, acute medication-use changes, treatment satisfaction, and safety.

Methods

Study Design

This was a simulated phase IIb, multicenter, randomized, double-blind, placebo-controlled, parallel-group study conducted for internal scientific-content workflow demonstration. The study included a 4-week baseline observation period followed by 16 weeks of randomized treatment.

Participants were randomized in a 1:1:1 ratio to receive one of the following once-daily oral regimens:

- Zefranova 30 mg.
- Zefranova 60 mg.
- Matched placebo.

Randomization was stratified by baseline monthly migraine-day frequency and prior exposure to two or more preventive-treatment classes. The study was designed to evaluate dose response, early onset, and tolerability patterns rather than to establish confirmatory regulatory evidence.

Patient Population

The simulated population included adults aged 18–65 years with chronic migraine and at least four migraine days per month during the baseline period. Participants were required to have a documented history of inadequate response, intolerance, or discontinuation of at least one prior preventive migraine therapy.

The population was designed to resemble patients frequently managed in specialist headache practice, including individuals with high baseline migraine frequency, frequent acute medication use, and clinically meaningful impairment in daily functioning.

Inclusion Criteria

Participants were eligible if they met the following simulated criteria:

- Age 18–65 years.
- Clinical history consistent with migraine for at least 12 months.
- Chronic migraine pattern during baseline observation.
- Four or more migraine days per month during baseline.
- Prior inadequate response, intolerance, or discontinuation of at least one standard preventive migraine therapy.
- Stable acute migraine medication pattern before randomization.
- Ability to complete electronic headache diaries and patient-reported outcome instruments.
- Willingness to comply with scheduled visits and study procedures.

Exclusion Criteria

Participants were excluded for any of the following simulated reasons:

- Unstable neurological disease other than migraine.
- Clinically significant uncontrolled cardiovascular, hepatic, renal, psychiatric, or systemic illness.
- Recent initiation or dose change of another migraine preventive therapy.
- Prior exposure to Zefranova.
- Unstable pattern of acute medication overuse judged likely to confound endpoint interpretation.
- Pregnancy, lactation, or unwillingness to follow pregnancy-prevention requirements during the study period.
- Participation in another interventional study within a prespecified washout period.
- Inability to complete electronic diary reporting reliably.

Study Treatments

Participants received once-daily oral Zefranova 30 mg, once-daily oral Zefranova 60 mg, or matched placebo for 16 weeks. Study medication was taken at approximately the same time each day. Use of permitted acute migraine medication was allowed and recorded in electronic diaries.

Concomitant initiation of new preventive migraine therapy was not permitted during the treatment period. Stable nonpharmacologic approaches, such as sleep-hygiene routines or previously established behavioral strategies, were allowed if unchanged during the study.

Outcome Measures

The primary endpoint was change from baseline in monthly migraine days during weeks 13–16.

Secondary endpoints included:

- Change from baseline in monthly migraine days at weeks 1–4, 5–8, and 9–12.
- Change from baseline in weekly migraine days during the first 4 weeks.
- Change from baseline in monthly headache days.
- Change from baseline in monthly acute medication-use days.
- Proportion of participants achieving $\geq 50\%$ reduction in monthly migraine days.
- Proportion of participants achieving $\geq 75\%$ reduction in monthly migraine days.
- Change from baseline in Headache Impact Test-6 score.
- Change from baseline in Migraine-Specific Quality of Life Role Function-Restrictive score.
- Patient-reported treatment satisfaction at week 16.
- Treatment adherence.
- Safety and tolerability outcomes.

Exploratory Responder Analysis

Exploratory analyses evaluated simulated baseline characteristics associated with $\geq 50\%$ migraine-day response at weeks 13–16. Variables included baseline monthly migraine-day frequency, baseline acute medication-use days, prior preventive-treatment class exposure, duration of migraine history, baseline headache-impact score, presence of sleep disturbance, and early response by week 4.

These analyses were descriptive and hypothesis-generating. No causal inference or validated predictive model is claimed.

Statistical Analysis

The simulated efficacy population included all randomized participants who received at least one dose of study medication and had baseline and post-baseline efficacy data. The safety population included all randomized participants who received at least one dose of study medication.

Continuous endpoints were analyzed descriptively using least-squares mean changes from baseline derived from a mixed model for repeated measures with treatment group, visit, baseline value, stratification factors, and treatment-by-visit interaction as model terms. Binary responder outcomes were summarized using counts, percentages, and descriptive risk differences. No regulatory-confirmatory statistical claim is made.

Missing data were handled using model-based estimation under a missing-at-random assumption, with sensitivity analyses using conservative imputation for participants who discontinued treatment.

Ethical and Data Considerations

This manuscript describes a simulated study and does not report data from an actual clinical trial. No real patients were enrolled, no real clinical sites participated, and no real ethics committee approval, informed consent process, investigator list, trial registry entry, or regulatory submission was involved. All study data, author names, institutions, and outcomes are fictional and are used only for structured scientific-content demonstration.

Results

Patient Disposition

A total of 811 simulated participants were screened. Of these, 624 met eligibility criteria and were randomized: 208 to placebo, 208 to Zefranova 30 mg, and 208 to Zefranova 60 mg. Overall, 585 participants completed the 16-week treatment period.

Completion rates were 94.2% in the placebo group, 94.7% in the Zefranova 30 mg group, and 92.3% in the Zefranova 60 mg group. Discontinuation due to adverse events occurred in 2.9% of placebo-treated participants, 3.8% of participants receiving Zefranova 30 mg, and 5.8% of participants receiving Zefranova 60 mg.

Baseline Characteristics

Baseline characteristics were generally balanced across groups. Mean baseline monthly migraine days ranged from 15.9 to 16.2 across treatment arms. Mean monthly acute medication-use days ranged from 11.5 to 11.9. More than half of participants had prior exposure to two or more preventive-treatment classes.

Table 1. Baseline Demographic and Clinical Characteristics

Characteristic	Placebo (n = 208)	Zefranova 30 mg (n = 208)	Zefranova 60 mg (n = 208)
Mean age, years	42.4	42.1	41.9
Female, n (%)	151 (72.6%)	149 (71.6%)	153 (73.6%)
Mean migraine history, years	13.7	13.5	13.8
Mean monthly migraine days	16.0	15.9	16.2
Mean monthly headache days	20.9	20.7	21.1
Mean monthly acute medication-use days	11.6	11.5	11.9
Prior exposure to ≥ 2 preventive classes, n (%)	113 (54.3%)	116 (55.8%)	115 (55.3%)
Baseline HIT-6 score, mean	64.5	64.2	64.8
Baseline MSQ-RFR score, mean	47.8	48.2	47.5
Sleep disturbance	72 (34.6%)	69 (33.2%)	74 (35.6%)

Characteristic	Placebo (n = 208)	Zefranova 30 mg (n = 208)	Zefranova 60 mg (n = 208)
reported, n (%)			

HIT-6 = Headache Impact Test-6; MSQ-RFR = Migraine-Specific Quality of Life Role Function-Restrictive domain.

Primary Endpoint

During weeks 13–16, Zefranova demonstrated a simulated dose-responsive reduction in monthly migraine days compared with placebo. Least-squares mean reductions were –3.2 days with placebo, –4.9 days with Zefranova 30 mg, and –6.1 days with Zefranova 60 mg.

The simulated between-group difference versus placebo was –1.7 monthly migraine days for Zefranova 30 mg and –2.9 monthly migraine days for Zefranova 60 mg. The 60 mg dose showed the largest reduction, while the 30 mg dose demonstrated intermediate benefit.

Table 2. Primary and Key Secondary Efficacy Outcomes at Weeks 13–16

Endpoint	Placebo (n = 208)	Zefranova 30 mg (n = 208)	Zefranova 60 mg (n = 208)
Change in monthly migraine days	–3.2	–4.9	–6.1
Difference vs placebo	—	–1.7	–2.9
Change in monthly headache days	–3.9	–5.5	–6.9
Change in monthly acute medication-use days	–2.2	–3.4	–4.3
≥50% migraine-day responder rate, n (%)	60 (28.8%)	83 (39.9%)	100 (48.1%)
≥75% migraine-day responder rate, n (%)	19 (9.1%)	31 (14.9%)	43 (20.7%)
Change in HIT-6 score	–4.8	–6.4	–8.1
Change in MSQ-RFR score	+9.6	+12.7	+16.5
Treatment satisfaction, n	88 (42.3%)	111 (53.4%)	131 (63.0%)

Endpoint (%)	Placebo (n = 208)	Zefranova 30 mg (n = 208)	Zefranova 60 mg (n = 208)
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Time Course and Early Onset of Effect

The simulated time-course analysis suggested earlier separation from placebo with Zefranova 60 mg than with Zefranova 30 mg. During weeks 1–4, monthly migraine-day reduction was –1.7 with placebo, –2.3 with Zefranova 30 mg, and –3.1 with Zefranova 60 mg. By weeks 5–8, both Zefranova doses showed greater reductions than placebo, with a more pronounced effect in the 60 mg group.

Table 3. Time Course of Monthly Migraine-Day Reduction

Time Period	Placebo	Zefranova 30 mg	Zefranova 60 mg
Weeks 1–4	–1.7	–2.3	–3.1
Weeks 5–8	–2.4	–3.7	–4.8
Weeks 9–12	–2.9	–4.4	–5.6
Weeks 13–16	–3.2	–4.9	–6.1

The early-onset pattern may be clinically relevant because patients and clinicians often reassess preventive therapy within the first several months. Early improvement, particularly by week 4 or week 8, may support adherence and treatment confidence, although absence of early response should not automatically preclude later benefit.

Responder Outcomes

At weeks 13–16, a $\geq 50\%$ reduction in monthly migraine days was observed in 28.8% of participants receiving placebo, 39.9% of participants receiving Zefranova 30 mg, and 48.1% of participants receiving Zefranova 60 mg. A $\geq 75\%$ reduction was observed in 9.1%, 14.9%, and 20.7% of participants, respectively.

Responder analyses suggested that participants with lower baseline acute medication-use days, shorter migraine history, and early week-4 improvement were more likely to achieve $\geq 50\%$ response by weeks 13–16. However, meaningful response was also observed among participants with longer disease duration and prior exposure to multiple preventive classes.

Table 4. Exploratory Characteristics Associated With $\geq 50\%$ Response at Weeks 13–16

Baseline or Early Characteristic	Responder Pattern Observed in Simulated Analysis
Early reduction by week 4	Strongest descriptive association with week 16 response
Lower baseline acute medication-use days	Associated with higher probability of response
Prior exposure to one preventive class	Numerically higher response than exposure to ≥ 2 classes
Shorter migraine history	Modestly higher response probability

Baseline or Early Characteristic	Responder Pattern Observed in Simulated Analysis
Lower baseline HIT-6 score	Slightly higher response probability
Sleep disturbance at baseline	Associated with lower response probability
High baseline monthly migraine days	Response still observed, but variability was greater

These findings are exploratory and should not be interpreted as validated predictors of treatment success. They may, however, inform future study designs that evaluate treatment-response patterns and patient stratification.

Acute Medication Use

Zefranova was associated with dose-related reductions in monthly acute medication-use days. At weeks 13–16, acute medication-use days decreased by 2.2 days with placebo, 3.4 days with Zefranova 30 mg, and 4.3 days with Zefranova 60 mg. Reduction in acute medication use was generally aligned with migraine-day reduction but was not identical across all participants.

This outcome is clinically relevant because frequent acute medication use may complicate chronic migraine management and may contribute to persistent headache burden in some patients. Reduced reliance on acute medication may also reflect improved disease stability and greater confidence in preventive therapy.

Patient-Reported Outcomes

Both Zefranova doses were associated with greater improvement in headache impact and migraine-specific role functioning than placebo. HIT-6 score decreased by 4.8 points with placebo, 6.4 points with Zefranova 30 mg, and 8.1 points with Zefranova 60 mg. MSQ-RFR score increased by 9.6 points, 12.7 points, and 16.5 points, respectively.

Treatment satisfaction was reported by 42.3% of placebo-treated participants, 53.4% of participants receiving Zefranova 30 mg, and 63.0% receiving Zefranova 60 mg. Participants reporting satisfaction generally had greater reductions in migraine days, lower acute medication-use days, and improved headache-impact scores.

Adherence and Treatment Exposure

Mean treatment adherence was high across all groups: 87.5% with placebo, 88.1% with Zefranova 30 mg, and 86.4% with Zefranova 60 mg. Participants receiving Zefranova 60 mg had a slightly higher rate of temporary interruption due to adverse events, but most resumed treatment.

The once-daily oral regimen supported consistent use in this simulated setting. However, adherence in clinical practice may be influenced by access, cost, expectations, comorbidities, follow-up intensity, and perceived benefit.

Safety and Tolerability

Treatment-emergent adverse events were reported in 49.5% of placebo-treated participants, 54.8% receiving Zefranova 30 mg, and 61.1% receiving Zefranova 60 mg. Most adverse events were mild or moderate. The most common events were nausea, fatigue, somnolence, dizziness, and dry mouth.

Table 5. Safety and Tolerability Summary

Safety Outcome	Placebo (n = 208)	Zefranova 30 mg (n = 208)	Zefranova 60 mg (n = 208)
Any treatment-emergent adverse event, n (%)	103 (49.5%)	114 (54.8%)	127 (61.1%)
Mild adverse event, n (%)	69 (33.2%)	74 (35.6%)	79 (38.0%)
Moderate adverse event, n (%)	31 (14.9%)	36 (17.3%)	42 (20.2%)
Severe adverse event, n (%)	3 (1.4%)	4 (1.9%)	6 (2.9%)
Serious adverse event, n (%)	2 (1.0%)	2 (1.0%)	3 (1.4%)
Discontinuation due to adverse event, n (%)	6 (2.9%)	8 (3.8%)	12 (5.8%)
Nausea, n (%)	14 (6.7%)	21 (10.1%)	29 (13.9%)
Fatigue, n (%)	13 (6.3%)	19 (9.1%)	26 (12.5%)
Somnolence, n (%)	9 (4.3%)	15 (7.2%)	22 (10.6%)
Dizziness, n (%)	10 (4.8%)	14 (6.7%)	20 (9.6%)
Dry mouth, n (%)	8 (3.8%)	13 (6.3%)	18 (8.7%)

No simulated pattern of clinically meaningful hepatotoxicity, severe hypersensitivity, major cardiovascular imbalance, or progressive neurological adverse-event signal was observed. The modestly higher adverse-event rate with Zefranova 60 mg should be weighed against the larger efficacy effect observed at that dose.

Discussion

This simulated phase IIb randomized study evaluated two once-daily oral doses of Zefranova in adults with chronic migraine and prior inadequate response to preventive therapy. The results demonstrated a dose-responsive reduction in monthly migraine days, with the 60 mg dose

showing the largest simulated treatment effect and the 30 mg dose showing moderate efficacy with a somewhat lower adverse-event burden.

The primary endpoint showed a reduction in monthly migraine days of −6.1 with Zefranova 60 mg and −4.9 with Zefranova 30 mg compared with −3.2 for placebo. The corresponding simulated between-group differences suggest that both doses may have clinically meaningful preventive activity, with stronger efficacy at the higher dose. In chronic migraine, a reduction of even two to three additional migraine days per month versus placebo may be important when accompanied by reduced headache impact, reduced acute medication use, and improved functioning.

Early onset of benefit is another clinically relevant observation. Zefranova 60 mg showed early separation from placebo by week 4, while Zefranova 30 mg demonstrated clearer separation by week 8. In specialist practice, early improvement can reinforce adherence and patient confidence. However, delayed response is also common in preventive migraine therapy, and treatment decisions should consider the full clinical context rather than relying only on very early response.

Responder analyses showed that nearly half of participants receiving Zefranova 60 mg achieved a $\geq 50\%$ reduction in monthly migraine days by weeks 13–16. A smaller but meaningful proportion achieved $\geq 75\%$ reduction. These responder outcomes may be particularly useful for clinical discussions because they translate group-level mean changes into patient-level benefit categories.

The exploratory responder analysis suggested several possible response-associated characteristics, including early improvement, lower acute medication-use burden, shorter migraine history, and lower baseline headache impact. These findings should be interpreted cautiously because the analysis was descriptive and simulated. Nevertheless, they reflect clinically plausible domains that may warrant evaluation in future prospective studies.

Reduction in acute medication use was an important secondary outcome. Zefranova 60 mg was associated with a 4.3-day reduction in monthly acute medication-use days, while Zefranova 30 mg was associated with a 3.4-day reduction. For neurologists and headache specialists, this outcome may be especially meaningful in patients with frequent acute medication use or concern for medication-overuse patterns. Reduced acute medication reliance may indicate better preventive control and potentially lower treatment complexity.

Patient-reported outcomes also improved in a dose-responsive manner. HIT-6 and MSQ-RFR changes suggested reduced headache impact and improved role functioning, respectively. These outcomes are important because migraine-related disability is not fully represented by frequency counts alone. Patients often value improvements in predictability, daily functioning, cognitive confidence, and reduced disruption to work or social activities.

Safety findings showed a modest dose-related increase in adverse events, especially nausea, fatigue, somnolence, dizziness, and dry mouth. Most events were mild or moderate, and discontinuation due to adverse events remained relatively low. The 60 mg dose showed the greatest efficacy but also a higher adverse-event burden than the 30 mg dose. This trade-off is central to dose selection and future clinical development.

Overall, the simulated evidence supports a potential development strategy in which Zefranova 60 mg is evaluated as the primary efficacy-oriented dose, while Zefranova 30 mg may be further studied as an alternative for patients who require a more conservative tolerability approach. Future studies would need to confirm dose response, define long-term safety, evaluate durability, and assess performance across clinically relevant subgroups.

Clinical Implications for Neurologists and Headache Specialists

This simulated study provides several practical insights for specialist care. First, a dose-responsive preventive-treatment profile may allow clinicians to balance efficacy and tolerability based on patient needs. A patient with severe migraine burden and strong preference for maximum migraine-day reduction may be more aligned with a higher-dose strategy, while a patient with tolerability concerns or comorbid fatigue may benefit from a more conservative starting approach.

Second, early response may help guide treatment conversations. Improvement within the first 4–8 weeks can support adherence and reinforce treatment expectations. However, absence of early response should be interpreted cautiously because some patients may require longer exposure before meaningful benefit emerges.

Third, acute medication-use reduction should be monitored alongside monthly migraine days. A patient who experiences fewer migraine days but continues frequent acute medication use may require additional counseling, medication review, or behavioral support. Conversely, reduced acute medication use may signal improved preventive stability even when migraine-day reduction is moderate.

Fourth, patient-reported outcomes should be integrated into clinical assessment. HIT-6, MSQ-RFR, treatment satisfaction, and patient global impression measures can help capture meaningful functional improvement that may not be fully reflected in headache-frequency metrics.

Finally, dose selection should remain individualized. The simulated results suggest that Zefranova 60 mg may provide stronger efficacy, while Zefranova 30 mg may offer a potentially more tolerable option for some patients. Prospective clinical evaluation would be required before any real clinical recommendations could be made.

Limitations

This manuscript has important limitations. All data are simulated and do not represent real clinical evidence. No real patients were enrolled, no actual clinical trial was conducted, and no regulatory conclusions can be drawn. The study design, endpoints, results, author names, institutions, and dataset are fictional and intended only for structured scientific-content workflow demonstration.

The 16-week treatment duration limits assessment of long-term durability, persistence, delayed adverse events, and sustained functional improvement. The simulated sample may not fully reflect the heterogeneity of real-world chronic migraine populations, including patients with complex psychiatric comorbidities, medication-overuse headache, polypharmacy, pregnancy considerations, older age, or limited healthcare access.

The responder analysis was exploratory, descriptive, and not based on a validated predictive model. Associations between baseline characteristics and response should not be interpreted as causal or clinically actionable. The study did not include an active comparator, so no conclusions can be made regarding comparative effectiveness versus established preventive therapies.

Finally, adherence and patient-reported outcomes in simulated studies may appear more consistent than in real-world settings. Future clinical research would require prospective data collection, validated instruments, representative populations, long-term follow-up, and rigorous safety monitoring.

Conclusion

In this simulated phase IIb randomized controlled study, once-daily oral Zefranova demonstrated a dose-responsive reduction in monthly migraine days among adults with chronic migraine and prior inadequate response to preventive therapy. The 60 mg dose showed the greatest simulated efficacy, including stronger early onset, higher responder rates, greater reduction in acute medication use, and larger improvements in patient-reported outcomes. The 30 mg dose demonstrated moderate benefit with a somewhat lower adverse-event burden.

The simulated tolerability profile was generally manageable, with mostly mild to moderate adverse events and a modest dose-related increase in nausea, fatigue, somnolence, dizziness, and dry mouth. These findings support further prospective evaluation of Zefranova dose selection, response timing, long-term safety, and individualized treatment strategies.

Because this manuscript is based entirely on simulated data, it should not be interpreted as real-world evidence of efficacy, safety, regulatory status, or clinical use. Confirmation in rigorously designed clinical trials and representative real-world evidence studies would be necessary.

Data Transparency Statement

All data presented in this manuscript are simulated for internal scientific-content workflow demonstration. No real patient records, clinical trial datasets, investigator-site data, ethics committee materials, trial registration records, regulatory documents, or healthcare databases were used. The dataset was created solely to support structured manuscript development for a simulated investigational therapeutic concept.

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Conflicts of Interest

All author names, institutional affiliations, and disclosure statements are fictional. No real financial relationships, employment relationships, advisory roles, honoraria, consulting fees, equity holdings, intellectual-property interests, or conflicts of interest are represented.

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